

Dear Hiring Committee,

I am writing to apply the Application Programmer Assoc position at the University of Michigan, CoE Robotics. My experience and technical skill set align well with many of the qualifications outlined in the job description, and most importantly, I see it as an exciting opportunity to continue learning and growing within a collaborative and mission-driven environment.

Last year, I made a huge contribution to my summer internship project at Information and Technology Services (ITS) of the University of Michigan, I developed the core Python program for pSSID, an open-source Wi-Fi monitoring system initiated at the university. This involved integrating it with the perfSONAR batch processor and managing Dockerized services to support client-server authentication between Filebeat and OpenSearch. These efforts require dedicated research and proficiency in provisioning automation using Ansible within enterprise-level networks, and Linux system administration on Raspberry Pi. This project highlights my ability to deliver end-to-end development—from design and implementation to deployment, integration, and testing.

To assist you identifying suitable candidate for this position, I would be pleased to provide my reference in support of my application. Edward Colone (epcjr@umich.edu) was my supervisor during my internship at ITS and can fully attest to my capabilities, work ethic and dedication to this career path.

Thank you for considering my application.

Sincerely,

Dianlu He

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dhe@emich.edu

EXPERIENCE

University of Michigan, Information and Technology Services

Ann Arbor, Michigan

Software Engineer Intern

May 2024 – Dec 2024

- Developed the core Python program for an open-source project, pSSID Wi-Fi monitoring system, which integrates perfSONAR batch processor, featuring a robust command line interface, dynamic content rendering using Jinja2, an efficient Cron scheduler, and reliable error handling.
- Performed Ansible provisioning in enterprise network, including bootstrapping, Ubuntu system administration and network namespace configuration, pSSID software deployment, and client server authentication.
- Experienced with configuring Raspberry Pi, establishing mTLS between filebeat and logstash, and creating data visualization using OpenSearch and Grafana.
- Demonstrated strong ability to handle workloads effectively and collaborate seamlessly with experts from Internet2 and Virginia Tech.

Eastern Michigan University, College of Engineering and Technology

Ypsilanti, Michigan

Graduate Assistant

Aug 2023 – Apr 2024

- Contributed to academic research projects on ROS robotics, including the development of a GUI controller using Vue.js and integrating YOLO object detection for enhanced automation.
- Prepared research posters and presented a robotic system on Detroit Auto Fair.
- Researched with Gazebo in simulating robotic systems and environments, focusing on evaluating sensor data integration, robotic navigation, and task execution in dynamic scenarios.

Guangzhou Juli Trading Co., Ltd

Guangzhou, China

Marketing Director

Feb 2012 – Feb 2018

- Managed digital marketing, content creation, and analytics to deliver auto parts marketing strategies.
- Leveraged data-driven insights using advanced analytics techniques to optimize customer segmentation, campaign performance, and market research.

Sanofi Pasteur Biological Products Co., Ltd.

Shenzhen, China

Technician

Mar 2009 – Jan 2012

- Performed process validation to ensure compliance with regulatory standards and product quality.
- Acted as critical personnel in the assembly and operation of filling machine for flu vaccine production.
- Composed detailed manufacturing protocols and documentation to streamline operations.

EDUCATION

Eastern Michigan University

M.S. in Computer Science

May 2022 – Dec 2024

South China Agricultural University

B.S. in Biotechnology

Sep 2004 – Jul 2008

SKILLS

Ansible, Bash, Python, Java, PHP, CSS, Vue.js, Django, Flask, MySQL, MongoDB, PostgreSQL, Data Analysis, Machine Learning (Computer vision, Neural Networks)